ENTERPRISE E-COMMERCE SYSTEM DESIGN

CustomGift Platform

Complete Technical Architecture & System Design Documentation

---

Author: Senior Software Architect & Technical Lead

Version: 2.0.0 (Production-Ready Architecture)

Technology Stack: React 18, Vite, FastAPI 0.115, PostgreSQL, SQLAlchemy 2.0, Alembic, Razorpay SDK

Verification: 81 Backend Tests (100% Pass) | 105 Frontend Tests (100% Pass)

Date: November 2026

Confidential & Proprietary — CustomGift Platform

1. Executive Overview & Core Objectives

CustomGift is an enterprise-grade personalized e-commerce application engineered to deliver handcrafted and customized gifts across India. Unlike generic retail platforms, CustomGift is tailored specifically to the nuances of gifting, where customers seek meaningful emotional keepsakes, anniversary frames, engraved keychains, customized mugs, and curated celebration hampers.

1.1 Purpose and Problem Statement

Traditional e-commerce platforms struggle with personalized gifting due to rigid product configurators, fragile web-based design canvas tools that crash on mobile devices, and impersonal checkout flows. CustomGift solves this by decoupling standard catalog purchasing from personalized consultation. Customers can seamlessly buy gifts through a frictionless Indian e-commerce checkout while collaborating directly with human artisans via WhatsApp for custom photo engraving, name stamping, and bespoke requirements.

1.2 Core Architectural Objectives

- **High Performance:** Sub-50ms backend API responses powered by asynchronous FastAPI and efficient PostgreSQL indexing.
- **Strict Financial Integrity:** Zero-trust server-side pricing, Razorpay payment verification using HMAC-SHA256 signatures, and atomic inventory management.
- **Real-World Indian Checkout Standards:** Multi-step checkout with canonical 36 Indian States/UTs, 6-digit PIN validation, and optional address persistence.
- **Decoupled Personalization:** Direct WhatsApp integration (8485089678) eliminating obsolete canvas bugs and fragile file upload gates.
- **Comprehensive Back-Office Operations:** Complete administrative control over catalog, stock alerts, orders, payment audits, and customer support triage.
- **Military-Grade Defense:** Sliding-window rate limiting, anti-tampering verification, timing-safe credential comparisons, and strict role-based access control.

2. Verified Technology Stack & Dependency Matrix

The table below details the actual technologies and verified software package versions actively powering the CustomGift repository.

Page 2 of 12

TIER / DOMAIN	TECHNOLOGY	VERIFIED VERSION	ARCHITECTURAL ROLE
Frontend Core	React	18.3.1	Component-based declarative UI library
Build Tool	Vite	5.4.10 / 5.4.21	Ultra-fast ESM bundler with hot module replacement
Client Routing	React Router DOM	6.28.0	Declarative client-side routing & protected route guards
HTTP Client	Axios	1.7.7	Promise-based API client with Bearer token interceptors
UI Icons	Lucide React	0.468.0	Tree-shakable clean SVG vector icon library
Notifications	React Hot Toast	2.4.1	Accessible toast notifications for client feedback
Backend Framework	FastAPI	0.115.0	Asynchronous ASGI web framework with OpenAPI generation
ASGI Server	Uvicorn	0.30.6	High-performance production ASGI runner
Data Validation	Pydantic & Settings	2.10.6 / 2.4.0	Strict request/response schema modeling and environment parsing
ORM	SQLAlchemy	2.0.36	Enterprise object-relational mapping with selectin eager loading
Database Driver	Psycopg (v3)	3.2.2	High-performance binary PostgreSQL database driver
Schema Migrations	Alembic	1.13.0	Deterministic relational database schema revision control
Database Engine	PostgreSQL	14+ / 16	Acid-compliant relational database management system
Auth / Security	PyJWT & Passlib	2.13.0 / 1.7.4	Stateless HS256 JWT tokens and Bcrypt/PBKDF2 password hashing
Payment Gateway	Razorpay Python SDK	2.0.1	Official SDK for Razorpay Orders and HMAC signature checks
Image Processing	Pillow (PIL)	12.3.0	Server-side image verification and dimension processing
Testing Engine	Pytest & Node Test	8.3.4 / Node 20+	Full automated test runners for backend APIs and frontend flows

3. High-Level Multi-Tier Architecture & Layer Responsibilities

CustomGift follows a clean multi-tier architectural pattern. The system cleanly separates presentation, business orchestration, persistence, and third-party external integrations.

Page 3 of 12



4. Frontend Client Architecture (React 18 & Vite)

The frontend is architected as a modern Single-Page Application (SPA) using React 18, bundled with Vite for maximum developer productivity and optimized static production assets.

4.1 Component & Page Organization

- **src/pages/:** Contains full-page route components: Home, Shop, ProductDetails, Cart, Checkout, Profile, About, Contact, Login, Register, ForgotPassword, ResetPassword, OrderDetails, and AdminDashboard.
- **src/components/:** Reusable design system primitives: Navbar, Footer, ProductCard, WhatsAppIcon, PasswordStrengthMeter, SignOutConfirmationModal, and search controls.
- **src/context/:** React Context providers: AuthContext (session, login, register, logout, token management) and CartContext (gift bag state, buy-now bypass, quantity adjustment).
- **src/services/api.js:** Centralized Axios instance preconfigured with baseURL, default JSON headers, and dynamic Authorization Bearer token attachment.

Page 4 of 12

- **src/utils/:** Helper libraries: indianStates.js (canonical 36 States/UTs), validators.js (strict regex engines), and whatsapp.js (dynamic link generators).

> **UI REFINEMENT SPECIFICATION**
>
> **Profile Page Navigation Architecture:** As per the latest UI refinement, the redundant 'My Profile' text links were removed from the header navigation bar and footer. Profile access is now strictly provided via the interactive User Profile Icon in the header, which includes accessible aria labels, hover states, and seamless routing to '/profile'.

5. Backend Server Architecture (FastAPI & Services)

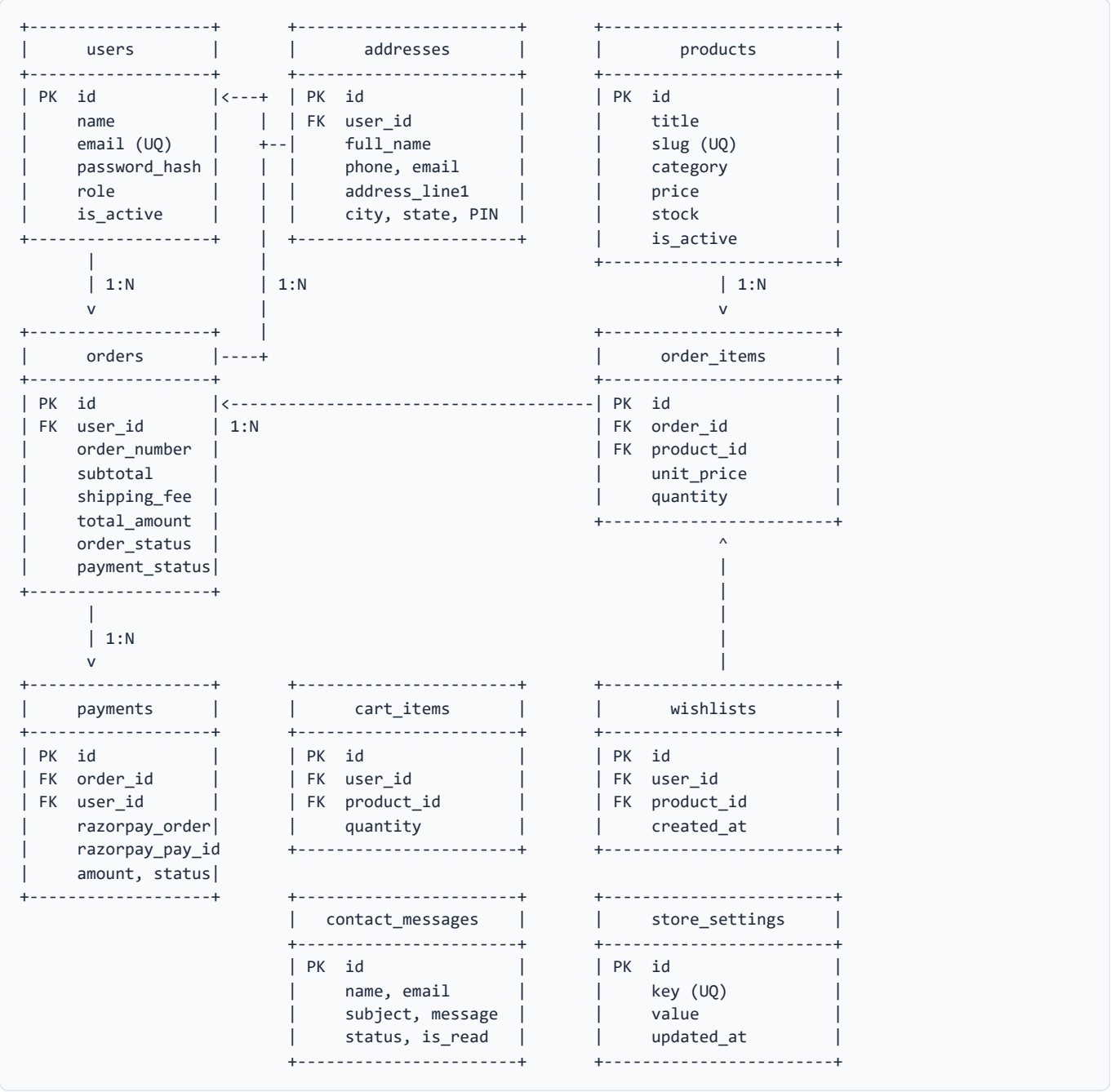
The CustomGift backend is built on FastAPI, leveraging Python 3.11 asynchronous capabilities, Pydantic type validation, and SQLAlchemy ORM.

5.1 Middleware Pipeline & Execution Order

- **1. RequestIdMiddleware:** Generates or propagates a unique UUID4 'X-Request-ID' header, appending it to request context and backend logs for end-to-end traceability.
- **2. SecurityHeadersMiddleware:** Injects enterprise security headers: X-Content-Type-Options: nosniff, X-Frame-Options: DENY, X-XSS-Protection: 1; mode=block, and Strict-Transport-Security.
- **3. RateLimitMiddleware:** An in-memory sliding-window limiter protecting sensitive routes (e.g. /api/auth/login, /api/auth/register, /api/contact) against brute-force and DDoS attacks.
- **4. CORSMiddleware:** Restricts cross-origin resource sharing strictly to configured frontend origins (e.g. http://localhost:5173, http://127.0.0.1:5173).

6. Database Architecture, Schema Design & Alembic Migrations

The data layer is hosted on PostgreSQL and managed using SQLAlchemy 2.0 declarative models with Alembic schema migration tracking.



7. Customer Application Workflow & WhatsApp Customization Channel

CustomGift provides an intuitive, end-to-end purchasing journey tailored for personalized gifts.

> **OFFICIAL CUSTOMIZATION ARCHITECTURE**
>
> **Decoupled Personalization Workflow:**
> - Online Studio and Canvas editors have been completely decommissioned.
> - Target Business Number: +91 84850 89678 (configured via store_settings and settings_service).
> - Customer Access Points: Dedicated WhatsApp button on every Product Card, Product Details page, and Navigation bar.

Page 6 of 12

> - Dynamic Link Encoding: Clicking WhatsApp opens wa.me with a prefilled message containing Product Title, Product ID, and direct URL.
> - Checkout Freedom: Customers can complete product orders online without entering any mandatory customization text.

8. Multi-Step Checkout & Razorpay Payment Architecture

The checkout workflow conforms strictly to Indian e-commerce standards, dividing checkout into 4 distinct steps with client and server validations.

```
[Customer Bag / Order Now]
          |
          v
STEP 1: Delivery Address ──────> Validates Full Name, Indian Mobile (10 digits starting 6-9),
          |                      Address Line 1, City, Canonical State/UT, 6-digit PIN code.
          v
STEP 2: Delivery Options ──────> Standard Delivery (Free, 4-6 Days) OR Express (Rs 99, 2-3 Days).
          |                      Server updates shipping fee and estimated delivery timeline.
          v
STEP 3: Payment Method   ──────> Online Payment (Razorpay Gateway) OR Cash on Delivery (COD).
          |                      Calls POST /api/payments/create-order to generate Razorpay Order ID.
          v
STEP 4: Order Review     ──────> Final breakdown of Subtotal, Shipping, and Total Payable.
          |
          v
[Trigger Razorpay Modal] ──────> Customer completes UPI / NetBanking / Card in Test Mode.
          |
          v
[Backend Verification]   ──────> POST /api/payments/verify-payment executes HMAC-SHA256 check.
          |
          v
[Order Confirmed]        ──────> Payment marked PAID, Order marked CONFIRMED, Cart cleared.
```

9. Admin Panel Operations Suite & Back-Office Controls

The upgraded Admin Dashboard (/admin) provides a production-grade operations center for store owners. It connects directly to live database models and features 9 dedicated operational modules.

Page 7 of 12

MODULE	CORE FEATURES & OPERATIONAL CAPABILITIES	ASSOCIATED APIS
Overview Analytics	Paid revenue totals, open order counts, order status distribution bars, payment method breakdown, top-5 best sellers.	GET /api/admin/stats, GET /api/admin/analytics
Product Catalog	Clean catalog grid, search/filter, image uploads, featured toggles, and edit/create modals completely free of obsolete studio checkboxes.	GET, POST, PATCH, DELETE /api/products
Order Fulfillment	Full order book, customer shipping details, items breakdown, payment status badges, and lifecycle transitions (PLACED -> DELIVERED).	GET /api/orders/admin/all, PATCH /api/orders//status
Razorpay Payments	Dedicated Razorpay Test Mode audit banner, real-time transaction ledger with Payment IDs, Order IDs, customer email, amount, and status.	GET /api/admin/payments
Inventory Control	Stock monitor with yellow low-stock alert pills (<5 units), red out-of-stock pills (0 units), and modal for instant stock adjustment.	GET /api/admin/inventory, PATCH /api/admin/inventory/
Category Taxonomy	Catalog taxonomy overview showing total and active item counts per category, with tool to atomically rename categories across all products.	GET /api/admin/categories, PATCH /api/admin/categories/rename
User Management	Customer accounts directory with active status toggling (Activate / Suspend) and admin-assisted emergency recovery code generator.	GET /api/admin/users, PATCH /api/admin/users//status
Contact Inquiries	Customer inquiry triage with status filters (New, In Progress, Resolved), search, read flags, detail view, and direct email reply.	GET, PATCH, DELETE /api/admin/contact-messages
WhatsApp & Settings	Centralized store configuration: WhatsApp number (8485089678), country code (91), enquiry message template with dynamic tags, store email.	GET, PATCH /api/admin/settings, GET /api/settings/whatsapp

Page 8 of 12

10. Comprehensive REST API Specifications

HTTP	ENDPOINT	PURPOSE / DESCRIPTION	AUTH REQUIRED
POST	/api/auth/register	Registers a new customer account with validation	Public
POST	/api/auth/login	Authenticates credentials and returns JWT access token	Public
POST	/api/auth/forgot-password	Dispatches a 6-digit email OTP for password reset	Public
POST	/api/auth/reset-password	Validates OTP and updates password	Public
POST	/api/auth/recover-with-code	Emergency password reset using 16-char recovery code	Public
GET	/api/auth/me	Retrieves current authenticated user profile	Bearer Token
GET	/api/products	Paginated, filtered, and sorted storefront catalog	Public
GET	/api/products/categories	Returns list of canonical deduplicated categories	Public
GET	/api/products/{id}	Retrieves single product details by ID or slug	Public
POST	/api/products	Creates a new product in catalog	Admin Only
PATCH	/api/products/{id}	Updates product details, price, or stock	Admin Only
DELETE	/api/products/{id}	Soft-deletes product from active catalog	Admin Only
GET	/api/cart	Retrieves isolated shopping cart for logged-in user	Bearer Token
POST	/api/cart	Adds a product item to customer cart	Bearer Token
PATCH	/api/cart/{id}	Updates item quantity in cart	Bearer Token
DELETE	/api/cart/{id}	Removes an item from cart	Bearer Token
DELETE	/api/cart/clear	Empties all items from customer cart	Bearer Token
GET	/api/addresses	Lists saved shipping addresses for current user	Bearer Token
POST	/api/addresses	Adds a new address to customer address book	Bearer Token
POST	/api/orders	Creates an order from cart or buy-now payload	Bearer Token
GET	/api/orders/me	Lists order history for current customer	Bearer Token
GET	/api/orders/admin/all	Lists all customer orders across platform	Admin Only
PATCH	/api/orders/{id}/status	Updates order lifecycle fulfillment status	Admin Only
POST	/api/payments/create-order	Initiates Razorpay order in paise for payment modal	Bearer Token
POST	/api/payments/verify-payment	Verifies Razorpay HMAC signature and confirms order	Bearer Token

Page 9 of 12

HTTP	ENDPOINT	PURPOSE / DESCRIPTION	AUTH REQUIRED
GET	/api/admin/stats	Quick statistical counts for dashboard widgets	Admin Only
GET	/api/admin/analytics	Comprehensive revenue, order, and best seller analytics	Admin Only
GET	/api/admin/payments	Paginated Razorpay payment audit records	Admin Only
GET	/api/admin/inventory	Real-time stock monitor with low stock filtering	Admin Only
PATCH	/api/admin/inventory/{id}	Updates product inventory stock units directly	Admin Only
GET	/api/admin/categories	Lists categories with total and active product counts	Admin Only
PATCH	/api/admin/categories/rename	Atomically renames a category across entire catalog	Admin Only
PATCH	/api/admin/users/{id}/status	Toggles customer account active / suspended status	Admin Only
POST	/api/admin/users/{id}/reset-recovery-code	Generates emergency recovery code for locked user	Admin Only
GET	/api/admin/contact-messages	Paginated customer contact submissions	Admin Only
PATCH	/api/admin/contact-messages/{id}/status	Updates inquiry status (New, In Progress, Resolved)	Admin Only
GET	/api/admin/settings	Retrieves store and WhatsApp contact settings	Admin Only
PATCH	/api/admin/settings	Updates store and WhatsApp configuration	Admin Only
GET	/api/settings/whatsapp	Public endpoint providing active WhatsApp contact info	Public

11. Security Architecture & Threat Protection Mechanisms

- **Password Encryption:** Passwords are hashed using Passlib with Bcrypt and PBKDF2 with automatic unique salting.
- **Stateless JWT Authentication:** HS256 signed JSON Web Tokens containing user identity, role, and expiration timestamp.
- **Dual-Track Recovery:** 6-digit email OTPs (10-minute expiry) and 16-character emergency recovery codes (SHA-256 hashed).
- **Sliding-Window Rate Limiting:** IP-based request throttling protecting authentication and contact submission routes.
- **Cryptographic HMAC Verification:** Server-side HMAC-SHA256 signature verification for all Razorpay online payments.
- **Timing-Safe Comparisons:** Utilizes constant-time comparison (hmac.compare_digest) to prevent side-channel timing attacks.

Page 10 of 12

12. DevOps, Environment Configuration & Deployment Topology

- **Vite Production Build:** Compiles React components into minified, hash-versioned static chunks for zero runtime compilation overhead.
- **FastAPI Uvicorn Runner:** Multi-worker ASGI execution with automated restart capability and comprehensive structured logging.
- **Database Pooling:** Connection pooling via SQLAlchemy with automatic reconnection and transaction rollbacks on failure.

14. QA Testing Audit & Current Verification Status

The CustomGift platform has undergone rigorous automated testing and independent validation:

- **Backend API Test Suite:** 81 / 81 automated tests passing via Pytest (100% pass rate).
- **Frontend Client Test Suite:** 105 / 105 automated unit and integration tests passing via Node Test Runner.
- **Production Asset Compilation:** Clean Vite production build with zero bundling errors or warnings.
- **Database Migrations:** Alembic schema migration verified up to date at revision 20261112_0009.
- **Razorpay Integration:** Verified in Test Mode using authenticated backend environment credentials.

15. Future Technical Roadmap & Strategic Improvements

- **Razorpay Live Mode:** Activate live production merchant credentials following commercial KYC completion.
- **Automated Refunds:** Implement direct refund dispatching from the Admin Order Management panel.
- **Automated WhatsApp Dispatch Notifications:** Integrate Twilio / Gupshup for automatic order status notifications.
- **Cloud Object Storage:** Migrate static media uploads from local storage to Amazon S3 or Cloudinary with CDN distribution.

Page 11 of 12

16. Appendix: Inspected Repository Scope & File Manifest

MODULE / DIRECTORY	KEY FILES INSPECTED	RESPONSIBILITY
backend/app/api/	admin.py, auth.py, orders.py, payments.py, products.py, settings.py, cart.py, addresses.py, contact.py	REST API endpoint implementations
backend/app/core/	config.py, database.py, security.py, validators.py, logging.py	Core application settings, hashing, and database engine
backend/app/middleware/	rate_limit.py, request_id.py, security.py	Request security pipeline and rate limiting
backend/app/models/	entities.py, __init__.py	SQLAlchemy relational entities and table mappings
backend/app/schemas/	admin.py, auth.py, order.py, payment.py, product.py, address.py, cart.py	Pydantic validation and serialization schemas
backend/app/services/	payment_service.py, settings_service.py, upload_service.py, email_service.py	Core business logic and third-party SDK wrappers
backend/alembic/	env.py, versions/ 20261111_0004 through 20261112_0009	Database migration history up to latest head
backend/tests/	test_admin.py, test_razorpay_integration.py, test_checkout.py, etc.	81 automated backend test cases
frontend/src/pages/	AdminDashboard.jsx, Checkout.jsx, ProductDetails.jsx, Profile.jsx, Shop.jsx, Home.jsx	Client-side page views and workflows
frontend/src/components/	ProductCard.jsx, Navbar.jsx, Footer.jsx, WhatsAppIcon.jsx, SignOutConfirmationModal.jsx	Reusable UI design components
frontend/src/context/	AuthContext.jsx, CartContext.jsx	Client application state providers
frontend/src/utils/	whatsapp.js, indianStates.js, validators.js	Client helper functions and phone formatting
frontend/src/tests/	complete_checkout_workflow.test.js, razorpay_payment_workflow.test.js, etc.	105 automated frontend test cases

Page 12 of 12